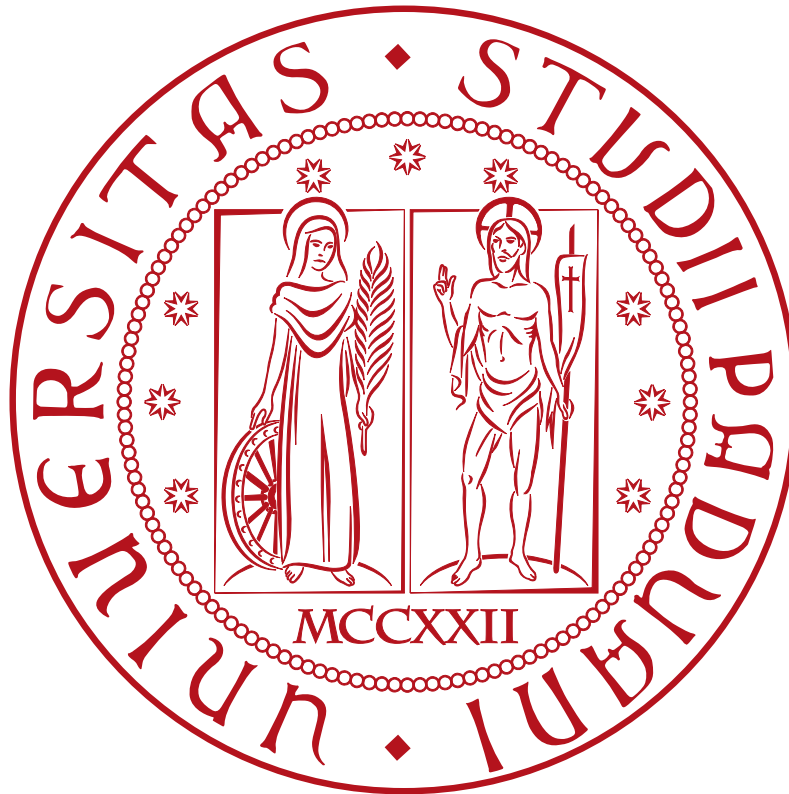


University of Padua



AIoT basics using Containerization

Exercises 1, 2, 3

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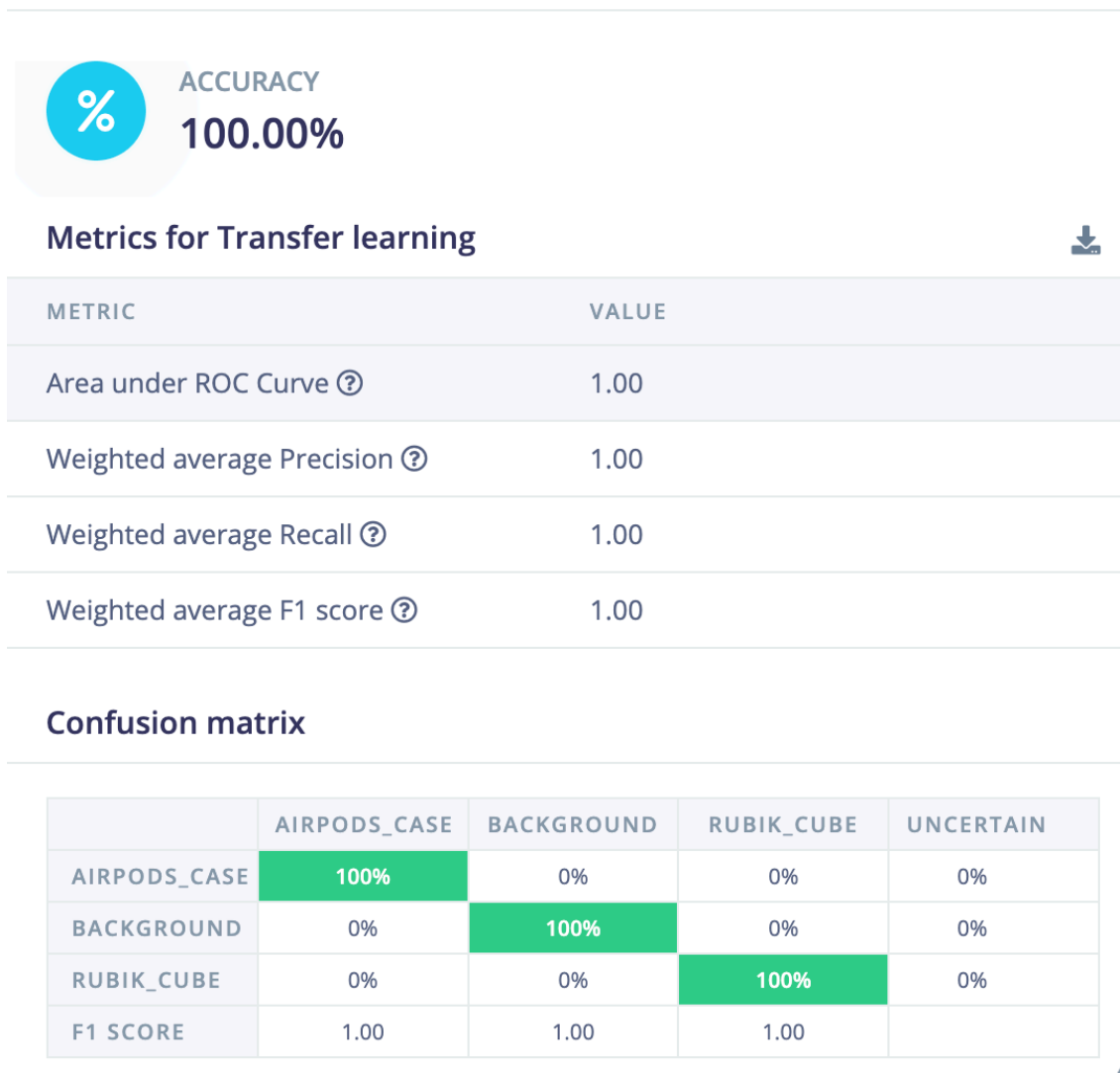
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1. Exercise 1

To make the first exercise I decided to classify 2 different items: a Rubik's cube and the AirPods' case. I took 40 pictures for each item, and also I took 40 pictures of the background.

After training the model with an 80/20 split, I got the following results in the model testing section of Edge Impulse:



Note that the accuracy is very high, but the dataset is very small and the task is very easy.

After the deployment here is the QR code to test the model on your phone:



Figure 1: QR code of the model

2. Exercise 2

For the second exercise, I built a dashboard to monitor some metrics of the CPU and memory usage of my device. In particular I used the plugins of Telegraf to collect the CPU and memory usage and I sent the data to InfluxDB. Then I created a dashboard in Grafana to visualize the data. All the components are running in Docker containers.

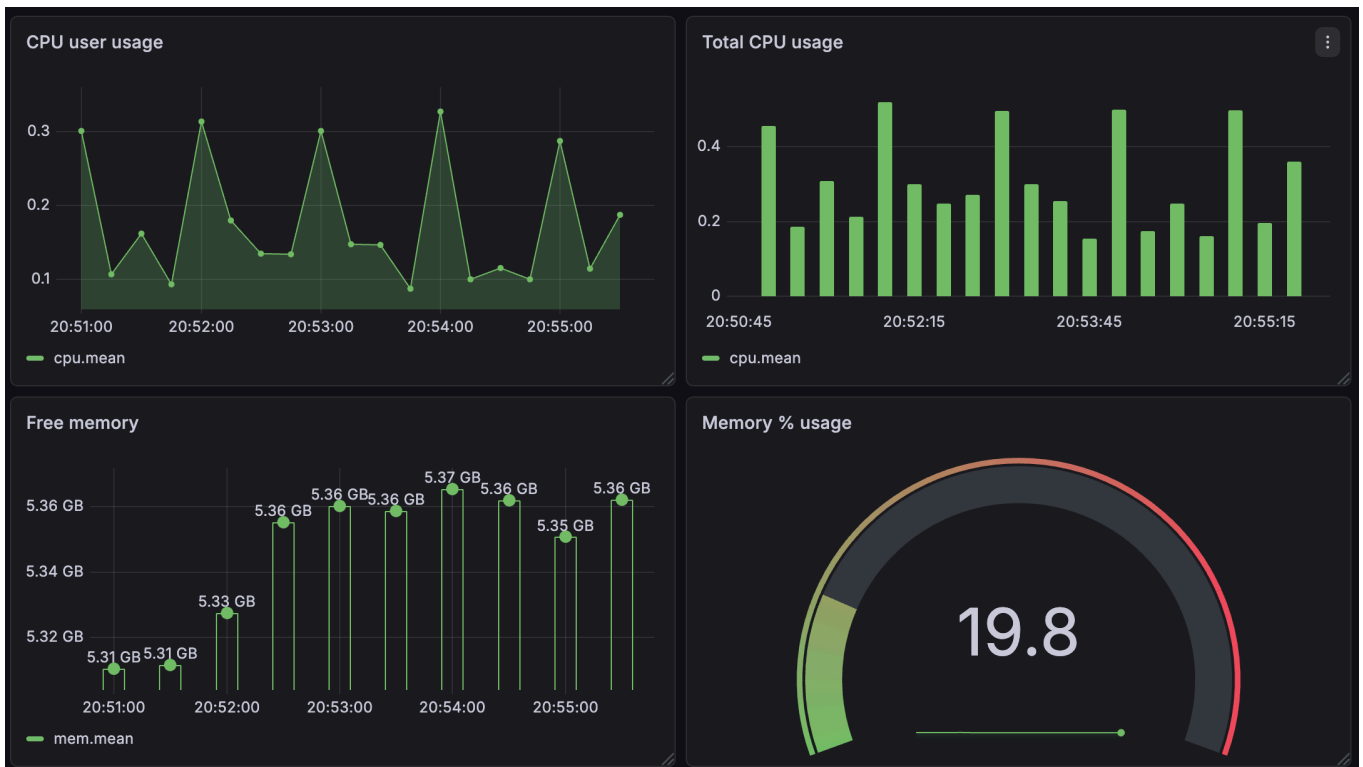


Figure 2: Grafana dashboard to monitor CPU and memory usage

The dashboard is very simple, on the upper left corner there is a graph that shows the CPU user usage over time, on the upper right corner there is a graph that shows the total CPU usage over time,

3. Exercise 3